Tianjian Li

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Education

Johns Hopkins University, Ph.D. in Computer Science — Aug 2024 - Present
- Advisor: Daniel Khashabi

Johns Hopkins University, M.S. in Computer Science — Aug 2022 - May 2024
- Advisor(s): Kenton Murray, Philipp Koehn, Daniel Khashabi

New York University, B.A. Joint Major in Mathematics and Computer Science — Aug 2017 - May 2021

Industry Experience

Research Scientist Intern, Meta Fundamental AI Research (FAIR) – New York, NY — June 2026 – Sept 2026
- Manager(s): Weizhe Yuan, Jason Weston.
- Topic: Monitoring reward hacking in auto-research agents.

Research Scientist Intern, Meta Fundamental AI Research (FAIR) – Bellevue, WA — May 2025 – Dec 2025
- Manager(s): Tianlu Wang, Jack Lanchantin, Jason Weston.
- Topic: Diversity / Exploration in RL post-training.

First-Authored Publications

Aligning Autoresearch: Co-Improvement between Agents and Humans [Paper][Blogpost][Dataset]
Tianjian Li, Chuanyang Jin, Seungone Kim, Tong Chen, John Hoffman, Yoram Bachrach, Pedro Rodriguez, Ping Yu, Eryk Helenowski, Swarnadeep Saha, Jack Lanchantin, Weizhe Yuan, Ilia Kulikov, Jason Weston.
Preprint 2026.

Self-Compacting Language Model Agents [Paper] [Code]
Tianjian Li, Jingyu Zhang, William Jurayj, Xi Wang, Chuanyang Jin, Mehrdad Farajtabar, Eric Nalisnick, Daniel Khashabi. NeurIPS 2026.

Jointly Reinforcing Diversity and Quality in Language Model Generations [Paper] [Code]
Tianjian Li, Yiming Zhang, Ping Yu, Swarnadeep Saha, Daniel Khashabi, Jason Weston, Jack Lanchantin, Tianlu Wang. NeurIPS 2026.

SimpleMix: Frustratingly Simple Mixing of Off- and On-policy Data in Language Model Preference Learning [Paper]
Tianjian Li, Daniel Khashabi. ICML 2025.

Upsample or Upweight: Balanced Training on Heavily Unbalanced Datasets [Paper][Video]
Tianjian Li, Weiting Tan, Haoran Xu, Kenton Murray, Daniel Khashabi. NAACL 2025.

Error Norm Truncation: Robust Training in the Presence of Data Noise in Text Generation Models [Paper][Code][Video]
Tianjian Li, Haoran Xu, Philipp Koehn, Daniel Khashabi, Kenton Murray.
ICLR 2024. Spotlight Presentation ($367/7304 \approx 5\%$).

Why Does Zero-shot Cross-lingual Transfer Fail? An Explanation and a Solution [Paper]
Tianjian Li, Kenton Murray. ACL 2023 Findings.

Non-First Authored Publications

ThoughtTrace: Understanding User Thoughts in Real-World LLM Interactions [Paper]
Chuanyang Jin, Binze Li, Haopeng Xie, Cathy Mengying Fang, **Tianjian Li**, Shayne Longpre, Hongxiang Gu, Maximillian Chen, Tianmin Shu.
RLxF Workshop, ICML 2026. Best Paper Award.

Many-Tier Instruction Hierarchy in LLM Agents [Paper]

Jingyu Zhang, **Tianjian Li**, William Jurayj, Hongyuan Zhan, Benjamin Van Durme, Daniel Khashabi.

EMNLP 2026 Findings.

Reasoning over Mathematical Objects: On-Policy Reward Modeling and Test Time Aggregation [Paper]

Pranjal Aggarwal, Marjan Ghazvininejad, Seungone Kim, Ilia Kulikov, Jack Lanchantin, Xian Li, **Tianjian Li***, Bo Liu, Graham Neubig, Anaelia Ovalle, Swarnadeep Saha, Sainbayar Sukhbaatar, Sean Welleck, Jason Weston, Chenxi Whitehouse, Adina Williams, Jing Xu, Ping Yu, Weizhe Yuan, Jingyu Zhang, Wenting Zhao. Preprint 2026.

**Alphabetical order. Led the section on parallel test-time scaling.*

The Flaw of Averages: Quantifying Uniformity of Performance on Benchmarks [Paper]

Arda Uzunoglu, **Tianjian Li**, Daniel Khashabi.

EMNLP 2026.

The Translation Barrier Hypothesis: Multilingual Generation with Large Language Models Suffers from Implicit Translation Failure [Paper]

Niyati Bafna, **Tianjian Li**, Kenton Murray, David R. Mortensen, David Yarowsky, Hale Sirin, Daniel Khashabi.

AACL 2025.

Benchmarking Language Model Creativity: A Case Study on Code Generation [Paper][Code][Video]

Yining Lu, Dixuan Wang, **Tianjian Li**, Dongwei Jiang, Sanjeev Khudanpur, Meng Jiang, Daniel Khashabi.

NAACL 2025.

Verifiable by Design: Aligning Language Models to Quote from Pre-Training Data [Paper][Code][Video]

Jingyu Zhang, Marc Marone, **Tianjian Li**, Benjamin Van Durme, Daniel Khashabi.

NAACL 2025. Oral Presentation.

Technologies

Languages: Python, C++, Java, Shell Script.

Frameworks and Tools: Linux, PyTorch, HuggingFace, Inference Engines (vLLM, SgLang), RL frameworks (verl, slime, prime-rl), Distributed Training (FSDP, Megatron-LM, DeepSpeed), vim, Latex.

Awards

National Olympiad in Informatics Provinces (NOIP) First Prize. 2015

Invited Talks

Beijing Academy of Artificial Intelligence (BAAI) Dec 2025

Johns Hopkins University, CLSP Seminar Spring 2024

Service

Reviewer for —

- NLP Conferences: ACL (2023, 2024, 2025), EMNLP (2023, 2024, 2025), NAACL (2025), EACL (2024, 2025), COLM (2024, 2025, 2026).
- ML Conferences: ICLR (2025, 2026), NeurIPS (2025), ICML (2026).

Organizer: Mid-Atlantic Student Colloquium on Speech, Language and Learning (2024)

Teaching

Head Teaching Assistant, JHU CS 601.471/671 NLP: Self-supervised Models Spring 2025

- Gave lectures on distributed training: [Video 1][Video 2]

Course Assistant, JHU CS 601.471/671 NLP: Self-supervised Models Spring 2024